

# Density Dependence

## Today's Agenda:

- Quiz
- Project 2: Species Interactions
  - Population growth
  - Density Dependence

## Project 2: Species Interactions: Due Feb 28

Step 1: Select type(s) of species interactions relevant for your study system

Step 2: Find and read 5-10 relevant papers

Step 3: Write around 3 pages about it (double spaced)

- ~ Pages 1-2: Focus on interactions

- ~ Page 3: Focus on impact of interactions on diversity patterns

Introduction format, references don't count against page limit

# Project 2: Species Interactions: Due Feb 28

Potential things to include:

- Predation
- Competition
- Mutualism
- Coexistence (with linkages to e.g., predation, competition, etc.)
- How things are tested
- Gaps in our knowledge
- Spatial variation in interaction strength

# Project 2: Species Interactions: Due Feb 28

## *Grade Breakdown:*

- At least five papers were referenced in the text 30%
- One or more interaction types were discussed 30%
- Impact of the interaction on diversity patterns was discussed 20%
- One or more additional aspects (e.g., testing, competition) discussed 10%
- Writing quality 10%
- Up to 10% will be deducted for excessive length -10%

Project 2: Species Interactions: Due Feb 28

Questions?

# Population Growth

# Population Growth

Thomas Malthus

- Economist who is critical for ecology and evolution



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- Focused on population growth





# Population Growth

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- Economist who is critical for ecology and evolution
- Focused on population growth
- Inspired the Exponential (a.k.a. Malthusian) Growth Model

$$N_t = N_0 e^{rt}$$



# Population Growth

## Thomas Malthus

- Economist who is critical for ecology and evolution
- Focused on population growth
- Inspired the Exponential (a.k.a. Malthusian) Growth Model

$$N_t = N_0 e^{rt}$$

$N_t$  = # at time  $t$ ,

$N_0$  = initial population size,

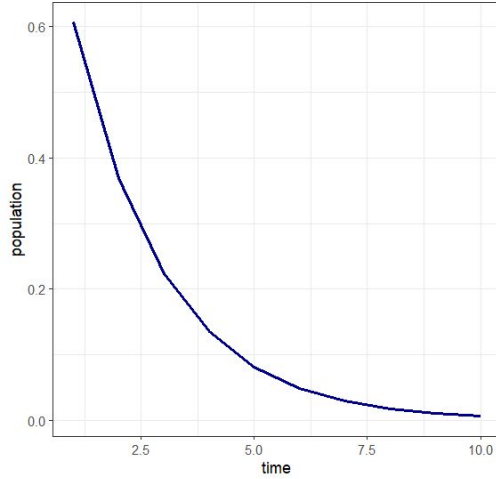
$r$  = intrinsic population growth rate

$t$  = time

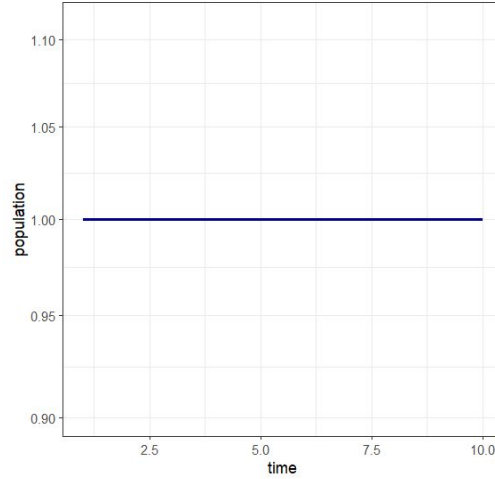


# Exponential Growth

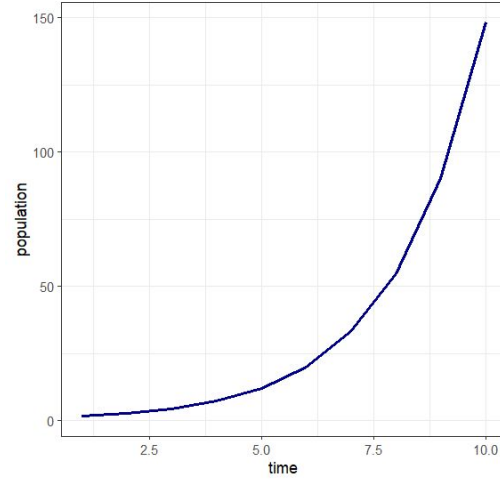
$$N_t = N_0 e^{rt}$$



$r = -0.5$



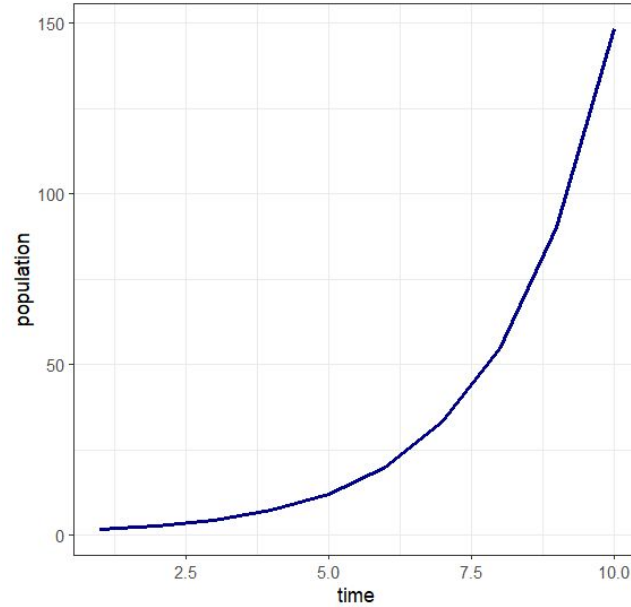
$r = 0.0$



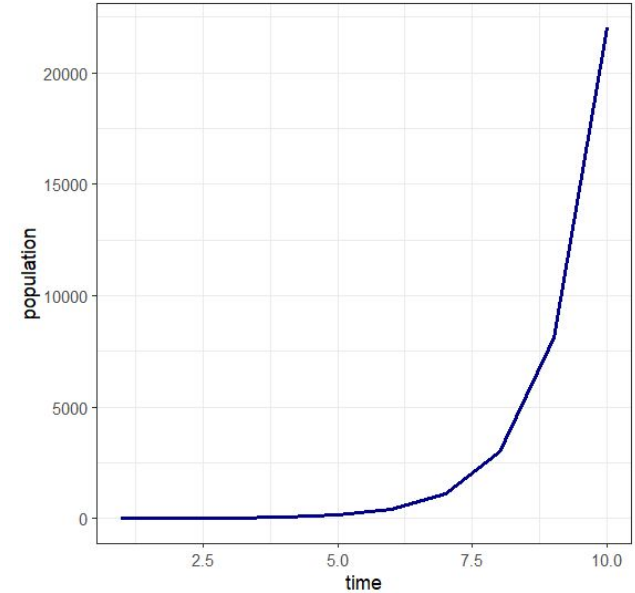
$r = 0.5$

# Exponential Growth

$$N_t = N_0 e^{rt}$$



$r = 0.5$



$r = 1$

# Exponential Growth

What is the growth rate for a population that doubles every 20 years?

$$N_t = N_0 e^{rt}$$

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$$2 = 1 * e^{r*20}$$

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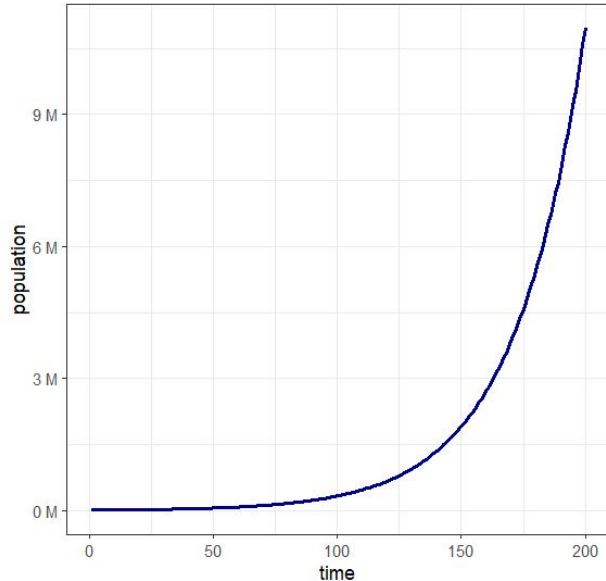
$$r = \ln(2)/20 = 0.03465736$$



# Exponential Growth

Our book gives 20,000 humans as an estimate for 70k bce

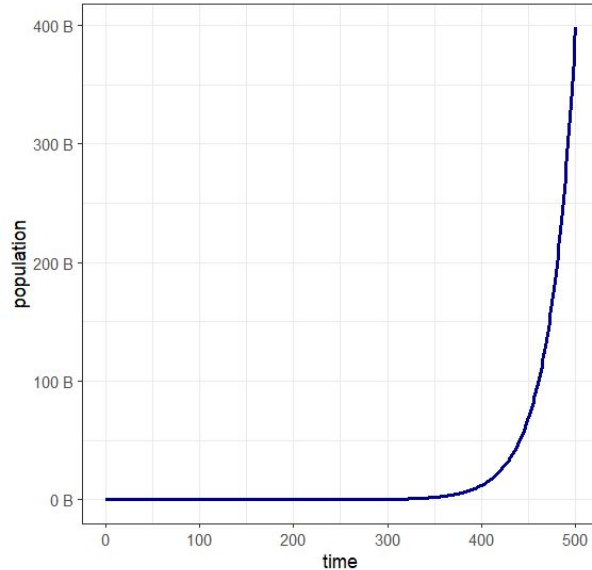
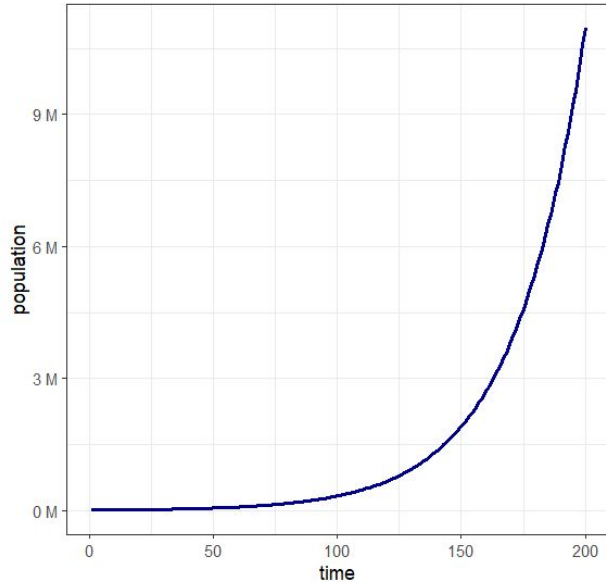
Assuming populations doubled in ~ 20 years...



# Exponential Growth

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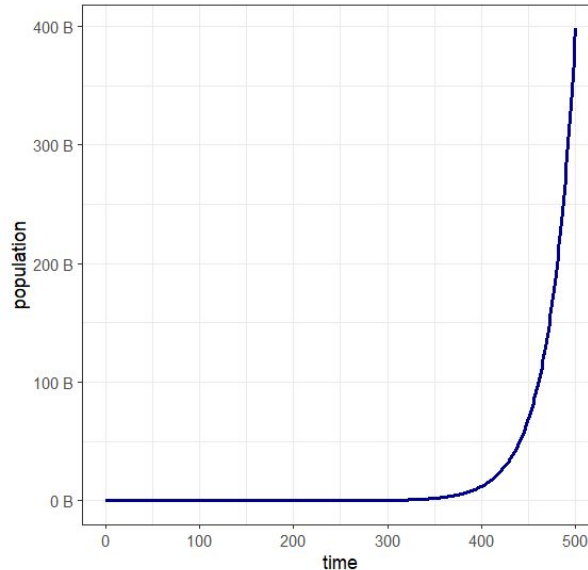
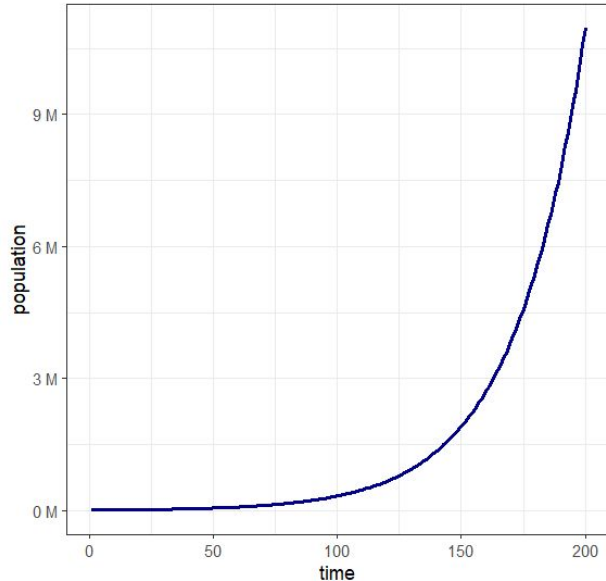
Assuming populations doubled in ~ 20 years...



# Exponential Growth

Our book gives 20,000 humans as an estimate for 70k bce

Assuming populations doubled in ~ 20 years...



Obviously, 400 billion humans is unrealistic...

# Density Dependence

As populations increase, growth rates decrease

What factors drive this?

# Density Dependence

As populations increase, growth rates decrease

What factors drive this?

- Competition
- Aggression
- Disease
- Predation
- Accumulation of waste

# Density Dependence

As populations increase, growth rates decrease

Creates a feedback between population size and growth rate

Often modelled using a logistic growth model

# Logistic Growth

$$N_t = N_0 \frac{K}{(K - N_0)e^{-rt} + N_0}$$

$N_t$  = # at time  $t$ ,

$N_0$  = initial population size,

$r$  = intrinsic population growth rate

$t$  = time

$K$  = carrying capacity

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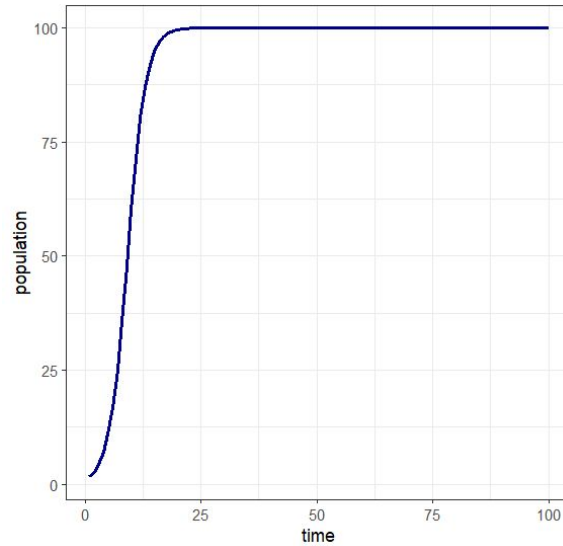
$K$  = carrying capacity



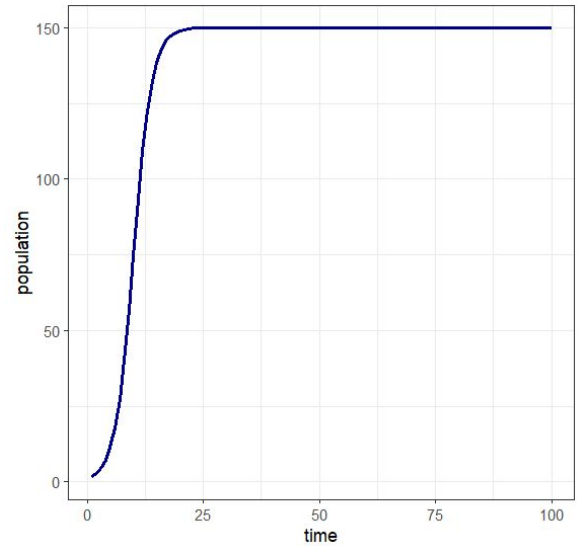
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$K = 100$   
 $r = 0.5$

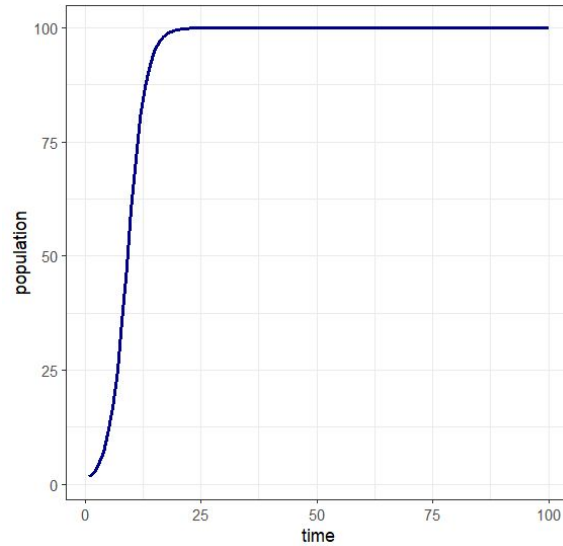


$K = 150$   
 $r = 0.5$

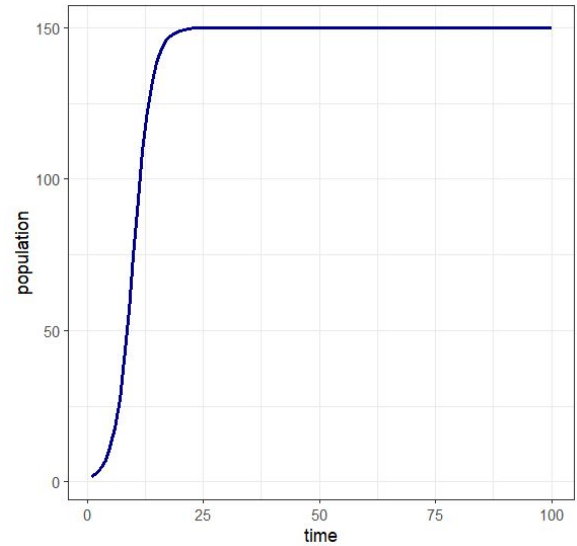
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 $t$  = time  
 $K$  = carrying capacity



$K = 100$   
 $r = 0.5$



$K = 150$   
 $r = 0.5$

$K$  controls maximum population size, that is,  
the carrying capacity

# Assumptions of Logistic Growth

Makes some assumptions, which are unlikely to hold:

- 1) Per capita growth rate is a linear function of the number of individuals
- 2) Changes to growth rate are instantaneous
- 3) External environment doesn't matter (e.g., resources, climate)
- 4) All individuals are equivalent

# Why use logistic growth?

Given the unrealistic assumptions, why use it?

- Good starting point
- Can add other features to it
- Often best to start with a simple model and only add parameters as needed
- Useful as a tool for thinking about population dynamics

# Causes of Density dependence

Earlier we discussed some factors that could lead to density dependence.

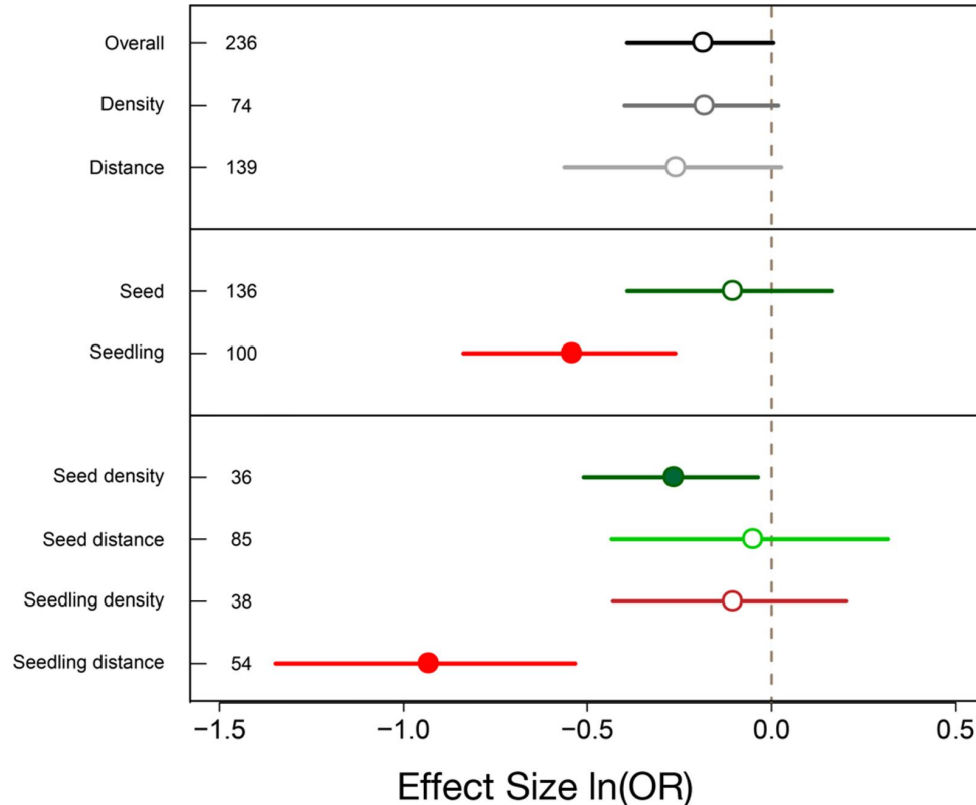
Lots of focus on competition

But, as we discussed, other factors can play a role!

# Janzen-Connell Hypothesis

- Focuses on specialist natural enemies (e.g., herbivores, pathogens)
- The impact of enemies is related to distance to others of your species
- So, as an area fills up, there are fewer places free of enemies
- Thus, populations are kept in check by enemies

# Janzen-Connell Hypothesis



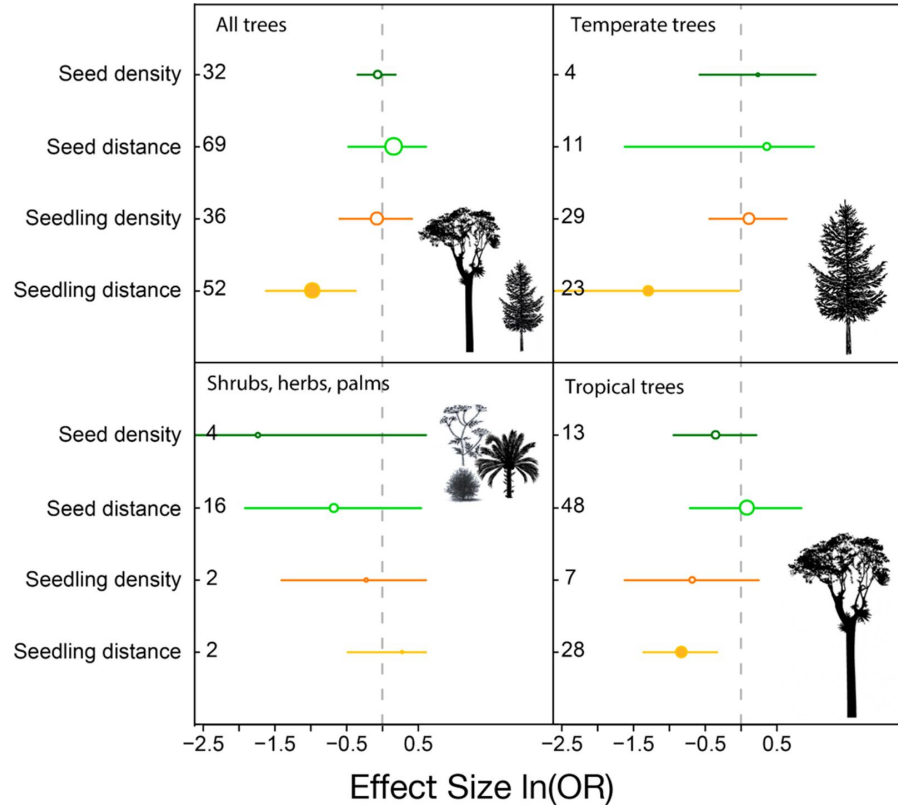
## ECOLOGY LETTERS

Review And Synthesis | [Full Access](#) |  

**When do Janzen–Connell effects matter? A phylogenetic meta-analysis of conspecific negative distance and density dependence experiments**

Xiaoyang Song, Jun Ying Lim, Jie Yang, Matthew Scott Luskin 

# Janzen-Connell Hypothesis



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# Positive density dependence?

Why might the growth rate of a species INCREASE with population size?

# Positive density dependence?

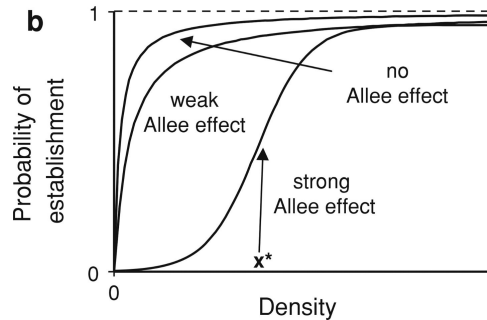
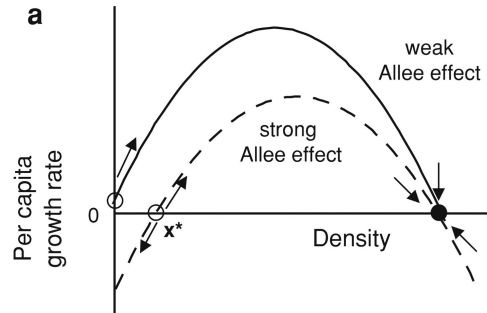
Why might the growth rate of a species INCREASE with population size?

- Ease of finding a mate
- Predator defense
- Offspring survival
- Greater foraging efficiency
- Environmental modification

Some of these are particularly important for social species

# Allee Effects

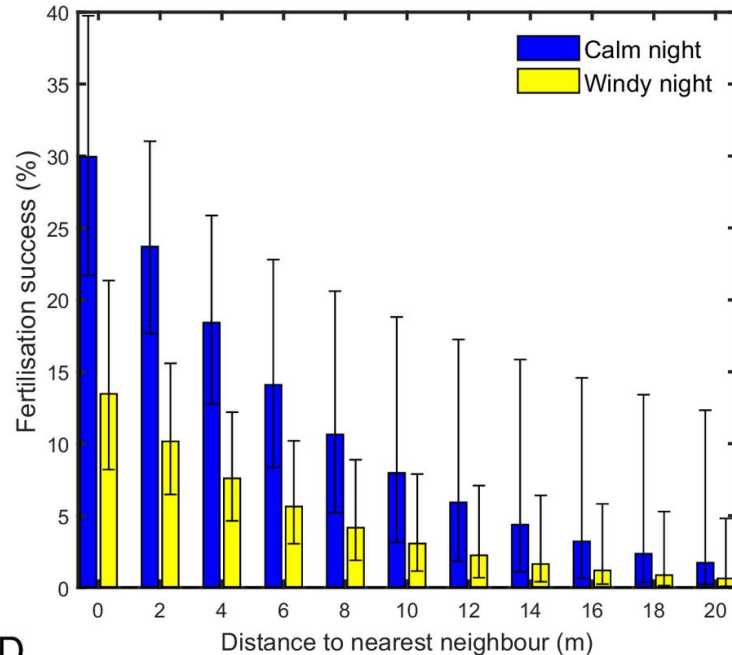
Positive relationship between growth rate and population size



# Allee Effects

Positive relationship between growth rate and population size

B



D

PNAS

RESEARCH ARTICLE | ECOLOGY

OPEN ACCESS

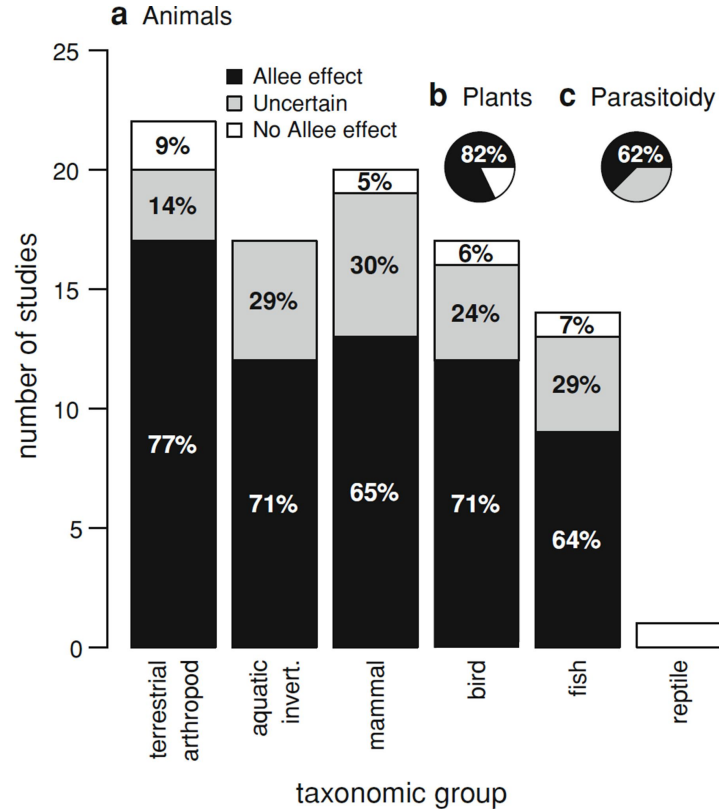
## Allee effects limit coral fertilization success

Peter J. Mumby <sup>a,b,1</sup>, Greta Sartori <sup>b</sup>, Elizabeth Burcher <sup>a,c</sup>, Cinzia Alessi <sup>b</sup>, Hannah Allan <sup>b</sup>,  
<sup>a,c</sup>, Christopher Doropoulos <sup>c</sup>, Geraldine Rengili <sup>b</sup>, and Gerard Ricardo <sup>a,c</sup>

Edited by Mark Hay, Georgia Institute of Technology, Atlanta, GA; received September 8, 2024; accepted October 24, 2024

December 16, 2024 | 121 (52) e2418314121 | <https://doi.org/10.1073/pnas.2418314121>

# Allee Effects



## Population Ecology

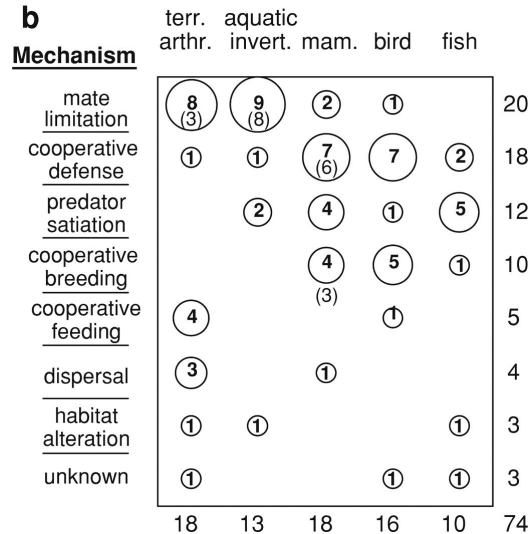
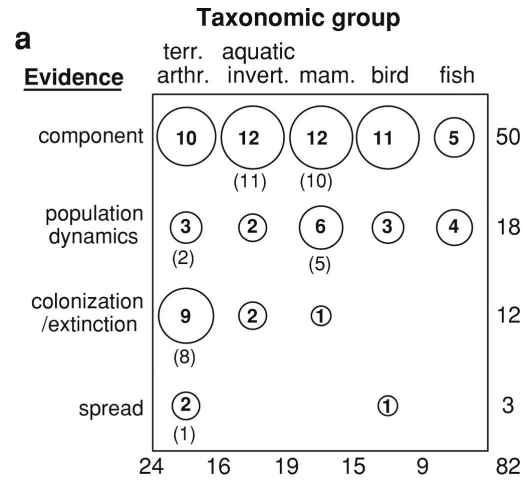
Special Feature: Review | [Full Access](#)

### The evidence for Allee effects

Andrew M. Kramer Brian Dennis, Andrew M. Liebhold, John M. Drake

First published: 18 April 2009 | <https://doi.org/10.1007/s10144-009-0152-6> | Citations: 384

# Allee Effects



## Population Ecology

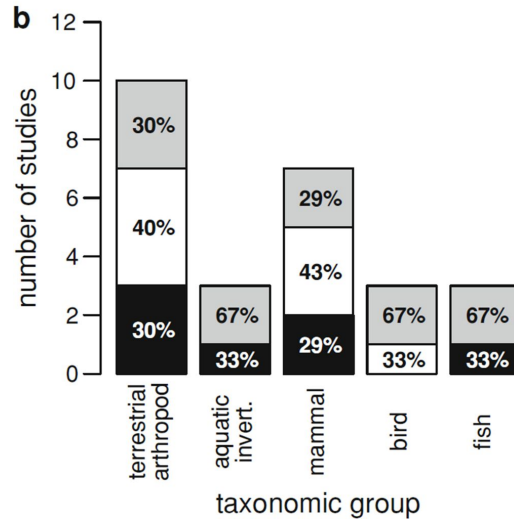
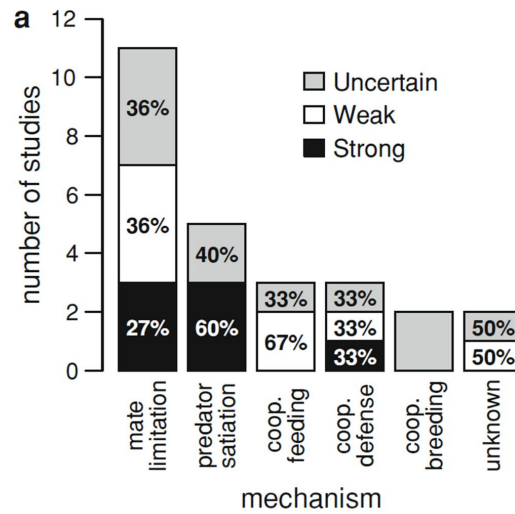
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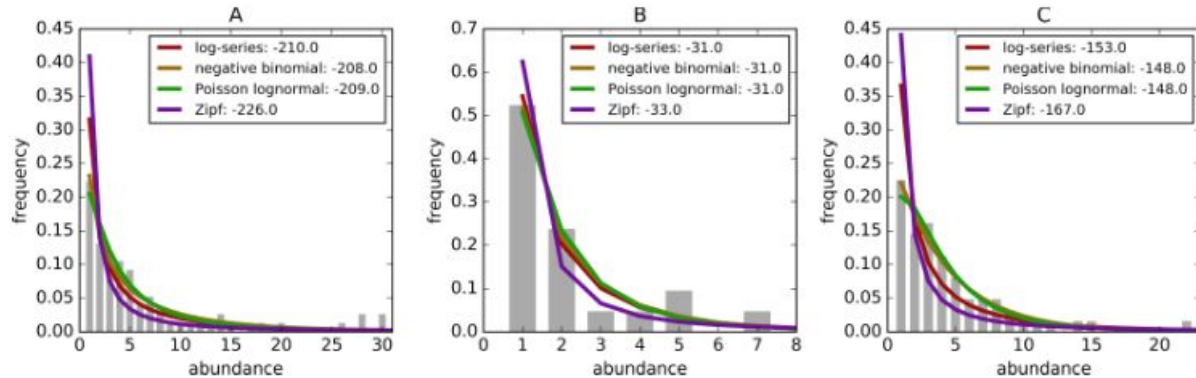
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# Density Dependence and Rarity

Remember SADs?



**Figure 1** Example species-abundance distributions including the empirical distributions (grey bars) and the best fitting log-series: maroon, negative binomial: brown, poisson lognormal: green, and Zipf: purple. Distributions are for (A) Breeding Bird Survey—Route 36 in New York, (B) Forest Inventory and Analysis—Unit 4, County 57, Plot 12 in Alabama, and (C) Gentry—Araracuara High Campina site in Colombia. Log-likelihoods of the models are included after the colon in the legend.

## An extensive comparison of species-abundance distribution models

Elita Baldrige<sup>1,2</sup>, David J. Harris<sup>3</sup>, Xiao Xiao<sup>1,2,4,5</sup> and Ethan P. White<sup>1,2,3,6</sup>

<sup>1</sup> Department of Biology, Utah State University, Logan, UT, United States

<sup>2</sup> Ecology Center, Utah State University, Logan, UT, United States

<sup>3</sup> Department of Wildlife Ecology and Conservation, University of Florida, Gainesville, FL, United States

<sup>4</sup> School of Biology and Ecology, University of Maine, Orono, ME, United States

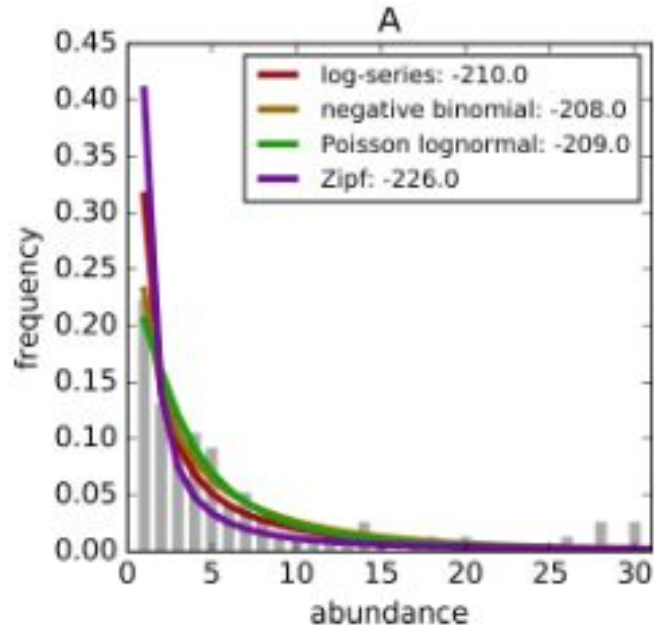
<sup>5</sup> Senator George J. Mitchell Center for Sustainability Solutions, University of Maine, Orono, ME, United States

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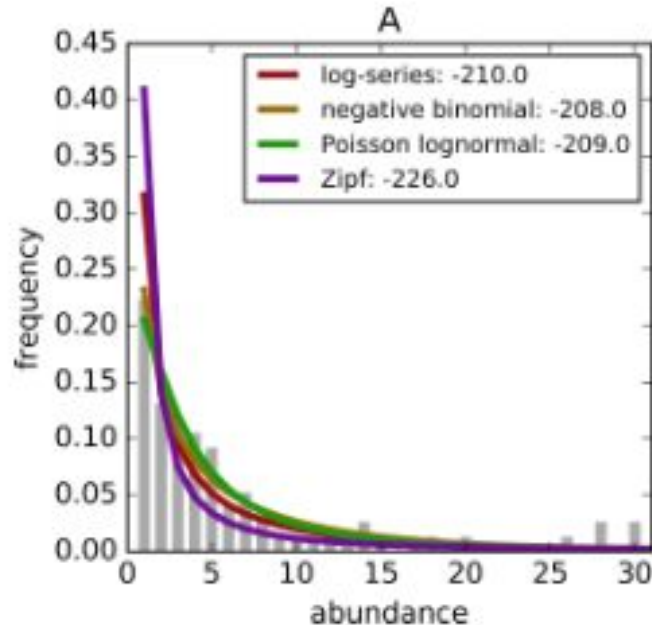
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Remember SADs?

Differences driven by density dependences?



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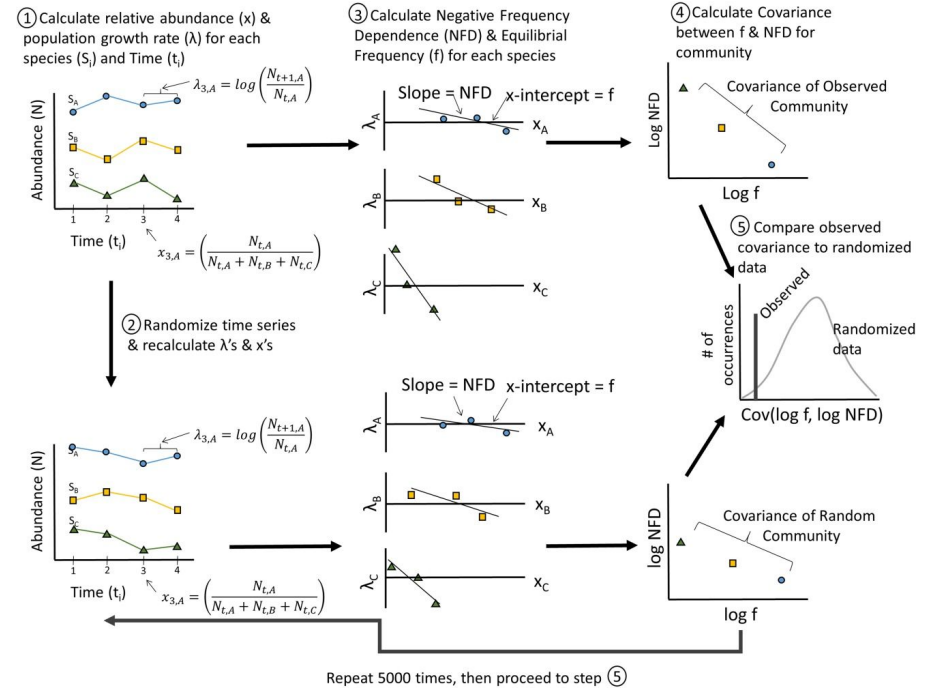
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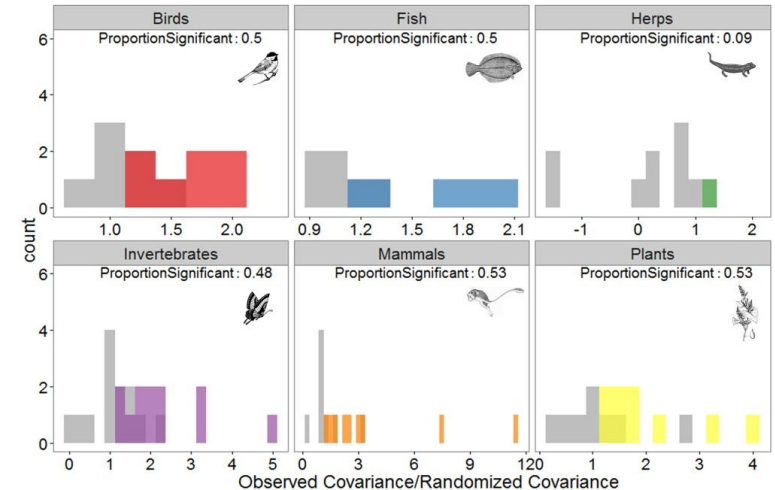
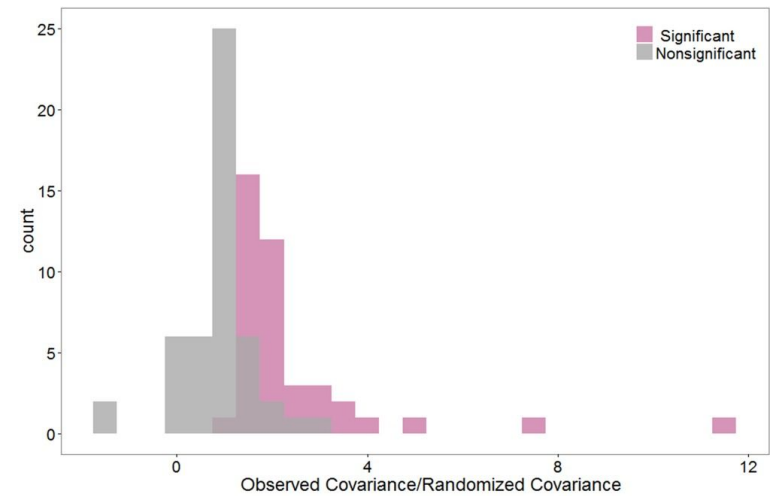
# Density Dependence and Rarity

- Rare species should experience stronger density dependence than common species



# Density Dependence and Rarity

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Global Ecology  
and Biogeography

A Journal of  
Macroecology

Concept Paper | [Full Access](#)

**Do persistent rare species experience stronger negative frequency dependence than common species?**

Glenda Yenni Peter B. Adler, S. K. Morgan Ernest

# Density Dependence and Rarity

Hypothesized that differences in density dependence may help buffer rare species

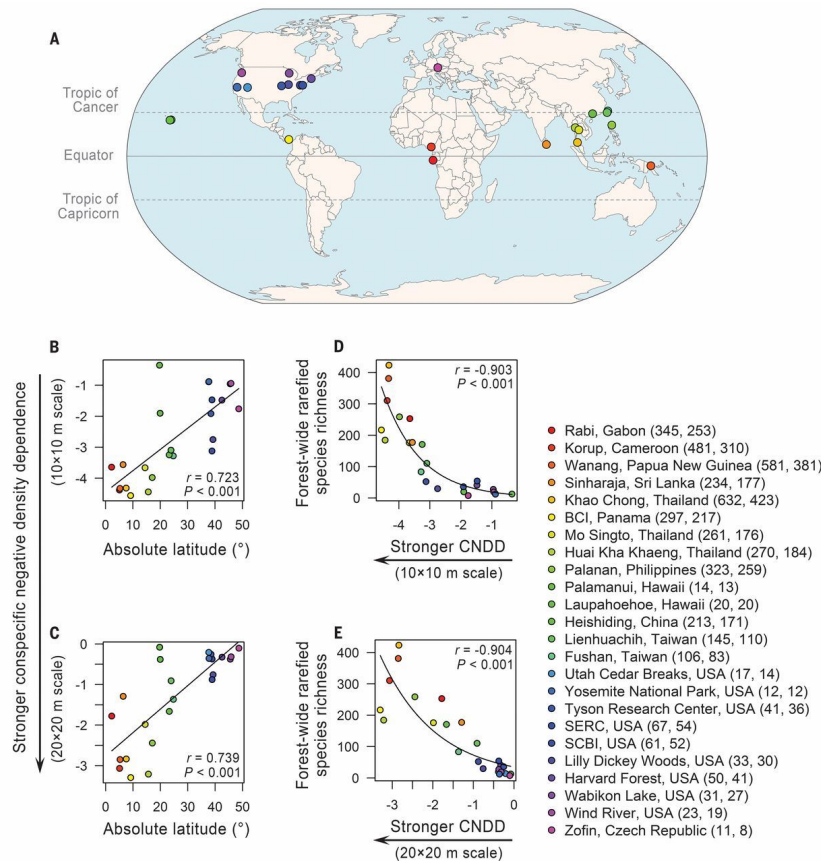
- Essentially improve growth rates when rare
- Small drop in population size is a big increase in growth rate

# Density dependence and richness

If there is variation in density dependence, does this impact richness?

- More DD, smaller populations
- Smaller populations, more species

# Density dependence and richness



## RESEARCH

Corrected 14 November 2018. See full text.

## FOREST ECOLOGY

# Plant diversity increases with the strength of negative density dependence at the global scale

Joseph A. LaManna,<sup>1,2\*</sup> Scott A. Mangan,<sup>2</sup> Alfonso Alonso,<sup>2</sup> Norman A. Bourg,<sup>4,5</sup> Warren Y. Brockelman,<sup>6,7</sup> Sarayudh Bunyavejchewin,<sup>8</sup> Li-Wan Chang,<sup>9</sup> Jyh-Min Chiang,<sup>10</sup> George B. Chuyong,<sup>11</sup> Keith Clay,<sup>12</sup> Richard Condit,<sup>13</sup> Susan Cordell,<sup>14</sup> Stuart J. Davies,<sup>15,16</sup> Tucker J. Furniss,<sup>17</sup> Christian P. Giardina,<sup>14</sup> I. A. U. Nimal Gunatilleke,<sup>18</sup> C. V. Savitri Gunatilleke,<sup>18</sup> Fangliang He,<sup>19,20</sup> Robert W. Howe,<sup>21</sup> Stephen P. Hubbell,<sup>22</sup> Chang-Fu Hsieh,<sup>23</sup> Faith M. Inman-Narahari,<sup>14</sup> David Janík,<sup>24</sup> Daniel J. Johnson,<sup>25</sup> David Kenfack,<sup>15,16</sup> Lisa Korte,<sup>3</sup> Kamil Král,<sup>24</sup> Andrew J. Larson,<sup>26</sup> James A. Lutz,<sup>17</sup> Sean M. McMahon,<sup>27,28</sup> William J. McShea,<sup>4</sup> Hervé R. Memiaghe,<sup>29</sup> Anuttara Nathalang,<sup>6</sup> Vojtech Novotny,<sup>30,31,32</sup> Perry S. Ong,<sup>33</sup> David A. Orwig,<sup>34</sup> Rebecca Ostertag,<sup>35</sup> Geoffrey G. Parker,<sup>28</sup> Richard P. Phillips,<sup>12</sup> Lawren Sack,<sup>22</sup> I-Fang Sun,<sup>36</sup> J. Sebastián Tello,<sup>37</sup> Duncan W. Thomas,<sup>38</sup> Benjamin L. Turner,<sup>13</sup> Dilys M. Vela Díaz,<sup>2</sup> Tomáš Vrška,<sup>24</sup> George D. Weiblen,<sup>39</sup> Amy Wolf,<sup>21,40</sup> Sandra Yap,<sup>41</sup> Jonathan A. Myers<sup>1,2</sup>

Next class: Predator-Prey Interactions